Differences in Differences

PS200C, Spring 2026

Chad Hazlett

UCLA

chazlett@ucla.edu

Roadmap

1. Theory: Potential outcomes, identification, key quantities
2. Randomization
   - difference in means, variance estimation
   - covariate adjustment
   - blocking
   - cluster randomization
3. Selection on Observables
   - sub-classification
   - matching (on X)
   - weighting (on X)
   - matching and weighting on $Pr(D = 1)$ (propensity scores)
   - regression
   - sensitivity
4. Difference in Differences
5. Instrumental Variables
6. Regression Discontinuity
7. More panel methods (time permitting)

Soo you don't buy SOO/CI, what next?

- A (relatively small) number of identification strategies have gained popularity as alternatives: DID, IV, RDD, and some time-series approaches
- These identification strategies are often associated with the "credibility revolution", but they are not "more causal" than SOO/CI – they just live on different assumptions.
- For any of these, you'll find there is some degree of inappropriate faith in these techniques because of this association
- RESIST! Understand and critique the assumptions.

Today: Difference-in-Differences (DID)

DID in the Time of Cholera

- Prevailing 19th century wisdom was that cholera was spread by “bad air”
- John Snow, a London physician thought otherwise
- In 1849, the Southwark and Vauxhall water company and the Lambeth company got water from the same polluted part of the Thames in central London. Snow assessed cholera mortality in both areas.
- In 1852, Lambeth moved its waterworks up stream.
- Shortly after, cholera death rates fell sharply in areas served by Lambeth, but not area served by S & V.

Intuitions:

- Why not just compare the two in 1852?
- Why not just compare Lambeth’s service area post-minus-pre?

Two Groups, Two Periods

Definition

Two groups:

- $D = 1$ for (eventually) treated units
- $D = 0$ for control units

Two periods:

- $T = 0$ Pre-Treatment period
- $T = 1$ Post-Treatment period

Potential outcomes $Y_{it}(d)$

- $Y_{it}(1)$: outcome unit i would attain in period t if treated by then
- $Y_{it}(0)$: outcome unit i would attain in period t if not treated

Two Groups and Two Periods

Definition

Causal effect for unit i at time t is

$$\tau_{it} = Y_{it}(1) - Y_{it}(0)$$

Observed outcomes Y_{it} are realized as

$$Y_{it} = Y_{it}(1) \cdot (D_i \cdot (t = 1)) + Y_{it}(0) \cdot (1 - D_i \cdot (t = 1))$$

Estimand (ATT post-treatment)

$$\tau_{ATT} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1]$$

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{i1}(1)|D_i = 1] - \mathbb{E}[Y_{i1}(0)|D_i = 1]$$

Observed Moments:

	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1)\|D = 1]$	$\mathbb{E}[Y_0(0)\|D = 1]$
Control D=0	$\mathbb{E}[Y_1(0)\|D = 0]$	$\mathbb{E}[Y_0(0)\|D = 0]$

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{i1}(1)|D_i = 1] - \mathbb{E}[Y_{i1}(0)|D_i = 1]$$

Observed Moments:

	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1)|D = 1]$	$\mathbb{E}[Y_0(0)|D = 1]$
Control D=0	$\mathbb{E}[Y_1(0)|D = 0]$	$\mathbb{E}[Y_0(0)|D = 0]$

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{i1}(1)|D_i = 1] - \mathbb{E}[Y_{i1}(0)|D_i = 1]$$

Observed Moments:

	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1)|D = 1]$	$\mathbb{E}[Y_0(0)|D = 1]$
Control D=0	$\mathbb{E}[Y_1(0)|D = 0]$	$\mathbb{E}[Y_0(0)|D = 0]$

Problem

Missing potential outcome: $\mathbb{E}[Y_1(0)|D = 1]$*, ie. what is the average non-treatment outcome in the post-treatment period, for the treated?*

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{i1}(1)|D_i = 1] - \mathbb{E}[Y_{i1}(0)|D_i = 1]$$

Observed Moments:

	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1)\|D = 1]$	$\mathbb{E}[Y_0(0)\|D = 1]$
Control D=0	$\mathbb{E}[Y_1(0)\|D = 0]$	$\mathbb{E}[Y_0(0)\|D = 0]$

ID Strategy 1 (not DID): Over-time ("Post-minus-Pre") Comparison

- Estimand: $\mathbb{E}[Y_1(1)|D = 1] - \mathbb{E}[Y_0(0)|D = 1]$
- Works if: $\mathbb{E}[Y_0(0)|D = 1] = \mathbb{E}[Y_1(0)|D = 1]$
- What does this assumption mean?

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{i1}(1)|D_i = 1] - \mathbb{E}[Y_{i1}(0)|D_i = 1]$$

Observed Moments:

	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1)\|D = 1]$	$\mathbb{E}[Y_0(0)\|D = 1]$
Control D=0	$\mathbb{E}[Y_1(0)\|D = 0]$	$\mathbb{E}[Y_0(0)\|D = 0]$

ID Strategy 2 (not DID): Cross-Sectional Comparison (in Post)

- Use: $\mathbb{E}[Y_1|D = 1] - \mathbb{E}[Y_1|D = 0]$
- Assumes: $\mathbb{E}[Y_1(0)|D = 0] = \mathbb{E}[Y_1(0)|D = 1]$
- What does this mean?

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{i1}(1)|D_i = 1] - \mathbb{E}[Y_{i1}(0)|D_i = 1]$$

Observed Moments:

	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1)\vert D = 1]$	$\mathbb{E}[Y_0(0)\vert D = 1]$
Control D=0	$\mathbb{E}[Y_1(0)\vert D = 0]$	$\mathbb{E}[Y_0(0)\vert D = 0]$

ID Strategy 3: Difference-in-Differences (DID)

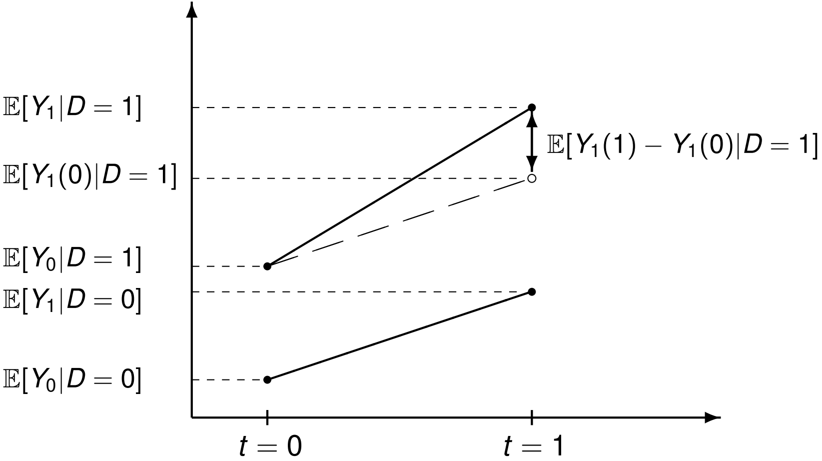
- Use:

$$\mathbb{E}[Y_1 - Y_0|D = 1] - \mathbb{E}[Y_1 - Y_0|D = 0]$$

- Assumes: $\mathbb{E}[Y_1(0) - Y_0(0)|D = 1] = \mathbb{E}[Y_1(0) - Y_0(0)|D = 0]$

"Parallel trends" assumption

Graphical Representation: Difference-in-Differences



Identification with Difference-in-Differences

Identification Assumption (Parallel Trends)

Conventionally we write,

$$\mathbb{E}[Y_1(0) - Y_0(0)|D = 1] = \mathbb{E}[Y_1(0) - Y_0(0)|D = 0]$$

But note that this is equivalent to:

$$\Delta Y(0) \perp D$$

where $\Delta Y(0) \equiv Y_1(0) - Y_0(0)$

Identfication Result

Under parallel trends, the ATT is identified by the DID estimator:

$$\begin{aligned}\tau_{DID} &= \{\mathbb{E}[Y_1|D = 1] - \mathbb{E}[Y_0|D = 1]\} - \{\mathbb{E}[Y_1|D = 0] - \mathbb{E}[Y_0|D = 0]\} \\ &= \mathbb{E}[Y_1(1) - Y_1(0)|D = 1] = ATT\end{aligned}$$

Proof

$$
\begin{aligned}
\tau_{DID} &= \{\mathbb{E}[Y_1|D=1] - \mathbb{E}[Y_0|D=1]\} - \{\mathbb{E}[Y_1|D=0] - \mathbb{E}[Y_0|D=0]\} \\
&= \{\mathbb{E}[Y_1(1)|D=1] - \mathbb{E}[Y_0(0)|D=1]\} - \{\mathbb{E}[Y_1(0)|D=0] - \mathbb{E}[Y_0(0)|D=0]\} \\
&= \{\mathbb{E}[Y_1(1)|D=1] - \mathbb{E}[Y_0(0)|D=1]\} - \{\mathbb{E}[Y_1(0)|D=1] - \mathbb{E}[Y_0(0)|D=1]\} \\
&= \mathbb{E}[Y_1(1)|D=1] - \mathbb{E}[Y_1(0)|D=1] \\
&= ATT
\end{aligned}
$$

Another view: additive effects

- Control group's observed change, $\mathbb{E}[\Delta Y \mid D=0]$, is "effect of time", δ_0.
- Treated group's observed change, $\mathbb{E}[\Delta Y \mid D=1]$, is effect of time plus treatment, $\delta_1 + \tau_{ATT}$.
- DID $= \mathbb{E}[\Delta Y \mid D=1] - \mathbb{E}[\Delta Y \mid D=0] = (\delta_1 + \tau_{ATT}) - \delta_0$.
- ...equals τ_{ATT} if $\delta_0 = \delta_1$ (parallel trends).

Why be skeptical of parallel trends?

If we're using DID, it's because we didn't trust the cross-sectional comparison:

$$Y(0) \not\perp\!\!\!\perp D$$

So why would we believe

$$\Delta Y(0) \perp\!\!\!\perp D \;?$$

Think of $\Delta Y(0)$ as a characteristic of each unit: how it would drift if untreated.

Is that characteristic really uncorrelated with who ends up treated?

Parallel trends isn't a *weaker* assumption—it's a *different* one. You need a story for why treatment assignment and unit-level drift are unrelated.

Ex ante reasons for hope or worry

Hopeful settings. Parallel trends is more plausible when the *selection mechanism* is blind to $Y(0)$ dynamics:

- **Decision lag**: treatment was decided well before it kicked in, weakening the relationship between who is treated and the trend come treatment time.
- **Exogenous trigger**: treatment assignment is tied to an abrupt external event (disaster, war, court ruling, sudden regulatory change), not to outcome trajectories. (Though it would have to still be related to $Y(0)$ to require DiD rather than straight cross-sectional comparison).
- **Administrative rollout**: decision is made at one level, then order is driven by logistics (alphabetical, pilot location, "who's ready first")...though if these are plausibly related to $Y(0)$, might be tough to argue they are unrelated to $\Delta Y(0)$.

Worrisome settings. Something *changing abruptly* around treatment time that shapes *both* who gets treated and the outcome:

- Treatment is a *response* to a recent shock or drift (Ashenfelter dip; programs target recent decliners).
- Latent forces influence both treatment taking (timing) and how $Y(0)$ would change absent treatment.

Difference-in-Differences: Estimators

Estimand (DID)

$$\tau_{DID} = \{\mathbb{E}[Y_1|D=1] - \mathbb{E}[Y_1|D=0]\} - \{\mathbb{E}[Y_0|D=1] - \mathbb{E}[Y_0|D=0]\}$$

Estimator

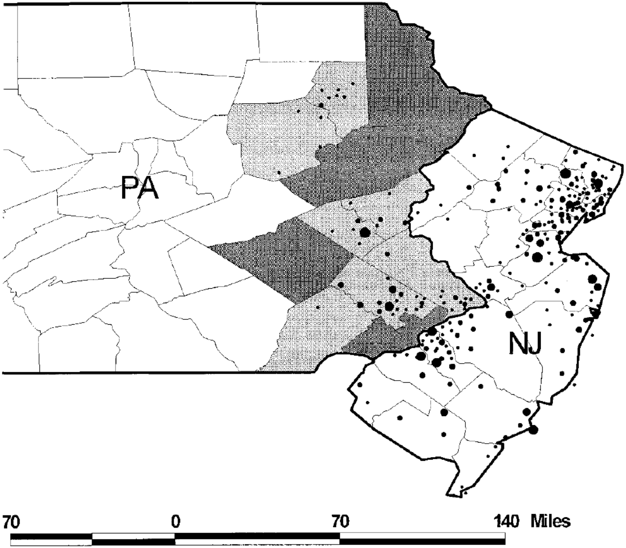
$$\frac{1}{N_1}\sum_{D_i=1}\{Y_{i1} - Y_{i0}\} - \frac{1}{N_0}\sum_{D_i=0}\{Y_{i1} - Y_{i0}\},$$

where N_1 and N_0 are the number of treated and control units respectively.

Example: Minimum wage laws and employment

- Do higher minimum wages decrease low-wage employment?
- Card and Krueger (1994) consider impact of New Jersey's 1992 minimum wage increase from \$4.25 to \$5.05 per hour
- Compare employment in 410 fast-food restaurants in New Jersey and eastern Pennsylvania before and after the rise
- Survey data on wages and employment from two waves:
  - Wave 1: March 1992, one month before min. wage increase
  - Wave 2: December 1992, eight months after increase

Locations of Restaurants (Card and Krueger 2000)



Sample Means: Minimum wage laws and employment

Variable	Stores by state PA (i)	NJ (ii)	Difference, NJ – PA (iii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)
3. Change in mean FTE employment	−2.16 (1.25)	0.59 (0.54)	2.76 (1.36)

Difference-in-Differences: Estimators

Estimator (Regression: Repeated Cross-Sections)

***In the two period case**, the same estimator can be computed by regression:*

$$Y = \mu + \gamma \cdot D + \delta \cdot T + \tau \cdot (D \cdot T) + \varepsilon,$$

where $\mathbb{E}[\varepsilon | D, T] = 0$.

	After (T=1)	**Before (T=0)**	**After - Before**
Treated D=1	$\mu + \gamma + \delta + \tau$	$\mu + \gamma$	$\delta + \tau$
Control D=0	$\mu + \delta$	μ	δ
Treated - Control	$\gamma + \tau$	γ	τ

Again, parallel trends: “time-effect” δ same for treated and controls

Regression: Minimum wage laws and employment

```
> d <- read.dta("CK1994_longformat.dta",convert.factors = FALSE)
> head(d[, c('ID', 'nj', 'postperiod', 'emptot')])
  ID nj postperiod emptot
1  1  0          0  40.50
2  1  0          1  24.00
3  2  0          0  13.75
4  2  0          1  11.50
5  3  0          0   8.50
6  3  0          1  10.50
```

Regression: Minimum wage laws and employment

You can reproduce the mean-differences estimator with a regression:

```
> ols <- lm(emptot ~ postperiod * nj, data = d)
> coeftest(ols)

t test of coefficients:

               Estimate Std. Error t value Pr(>|t|)
(Intercept)     23.3312     1.0719 21.7668  < 2e-16 ***
postperiod      -2.1656     1.5159 -1.4286  0.15351
nj              -2.8918     1.1935 -2.4229  0.01562 *
postperiod:nj    2.7536     1.6884  1.6309  0.10331
```

Three famous DiD studies: is parallel trends credible?

For each: imagine a *time-varying* confounder. What might make $\Delta Y(0)$ differ between treated and control?

1. **Card & Krueger (1994)**: NJ raises its minimum wage; PA doesn't.
   Y = fast-food employment. Compare NJ vs PA, before/after.
2. **Ashenfelter (1978)**: federal job-training program (MDTA/CETA). Trainees enroll; non-enrollees serve as controls.
   Y = earnings. Compare trainees vs non-trainees, before vs after enrollment.
3. **Autor (2003)**: US state courts recognized exceptions to the "employment-at-will" doctrine (make workers harder to fire) in different years. You look at whether there is more reliance on temporary workers (easier to fire).
4. Snow's cholera study: Lambeth (moves water works) vs. S&V (does not move water works), look at cholera mortality.

Some sneaky ways parallel trends can be violated

1. **Ashenfelter dip / mean reversion**: treatment is a response to a recent bad draw (people enroll in a training program *because* earnings just dipped; NGOs target villages that had a setback). The treated group is poised to recover even without treatment—so $\Delta Y(0)$ differs.

2. **Compositional change**: with repeated cross-sections, the treated and control samples aren't the same units over time. If treatment triggers migration or selective exit, the sampled pool shifts across periods and observed ΔY no longer estimates $\Delta Y(0)$.

3. **Short window vs long-run relevance**: parallel trends is much more plausible over short horizons. Over long ones, too much else happens—and short-term effects generally don't extrapolate to long-term ones.

4. **Functional-form dependence**: differencing is not scale-invariant. If outcomes grow at rates that depend on their *levels*, trends won't be parallel even with purely static confounding. Parallel in levels $\neq$ parallel in logs.

Checks for Difference-in-Differences Design

Is the parallel-trends assumption testable?

Checks for Difference-in-Differences Design

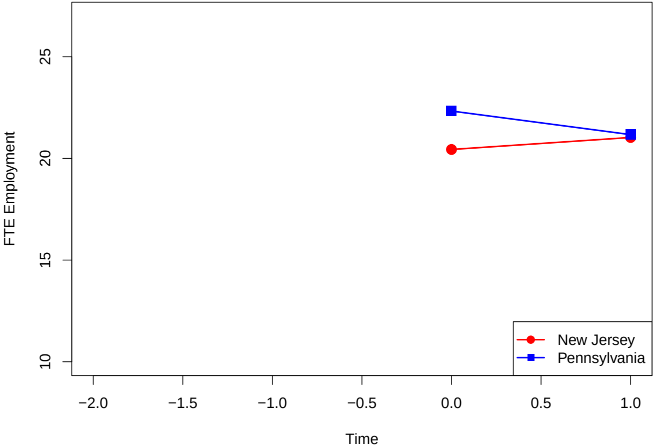
Two kinds of falsification tests for non-parallel trends

1. Falsification test using data for prior periods
   - Given parallel trends during periods $t = -1, 0, 1$, we have:

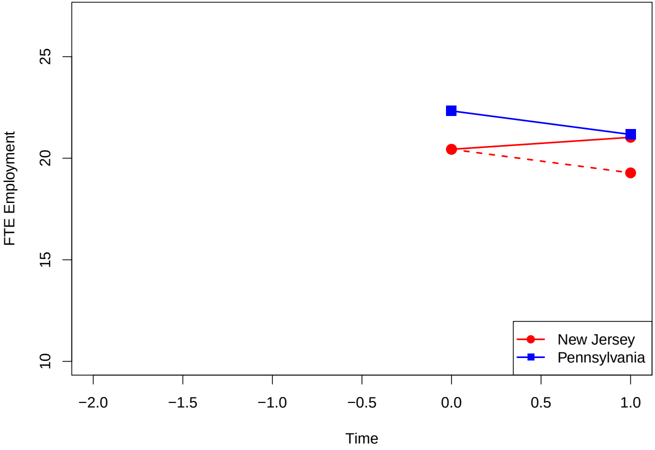
$$\mathbb{E}[Y_0 - Y_{-1}|D = 1] - \mathbb{E}[Y_0 - Y_{-1}|D = 0] = 0$$

   - run placebo DID on data from $t = -1, 0$.
   - what "treatment effect" should we get?
2. Falsification test using alternative outcome or treated sub-group that is not supposed to be affected by the treatment
   - troubling if DID from the placebo outcome or un-movable group is non-zero.

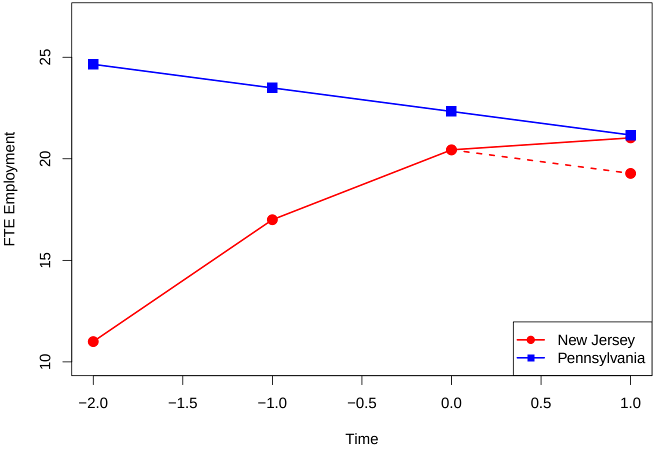
Falsification test: Data for prior periods



Falsification test: Data for prior periods

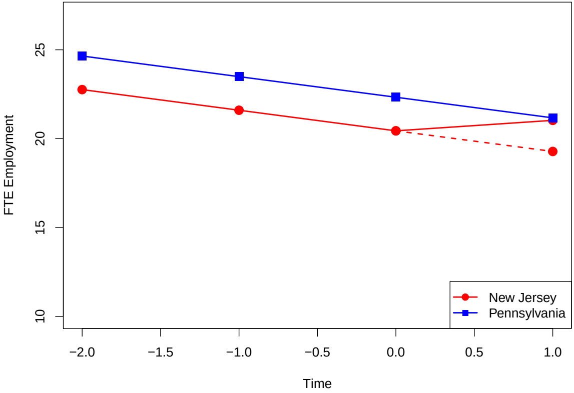


Falsification test: Data for prior periods



Definitely bad.

Falsification test: Data for prior periods



Not definitely bad, but why did they take treatment then?
Parallel pre-trends $\neq$ parallel trends.
You cannot test parallel trends.

Other approaches to worrying about parallel trends

1. **Sensitivity/ partial ID**
   - Best-known is perhaps Rambachan & Roth (2023)
   - Suggests checking if the effect estimate can survive non-parallelness k times larger than the non-parallelness in the prior trends.
   - A lot of papers don't even survive at $k = 1$, which is clearly bad (Chiu, Lan, Liu, Xu, *APSR* 2025)
   - Of course there is no (convincing) rule for "how big of a k is big enough"

We won't cover as much as I like but there are other ID opportunities for similar settings, and other sensitivity analyses depending on what you are prepared to reason about.

Vote in the comments to affect the story.

“Modern” DID?

- Classic DID above was two groups, two periods. Often you have units moving into treatment at different times (“staggered adoption”), and you’d still like to rely on something like parallel trends.
- People long used “two-way fixed effect” regressions in these cases and unfortunately called this DID.
- The trouble with TWFE is the combined result of:
  - **Already-treated units end up as “controls”** for later cohorts’ comparisons (mechanical; peculiar to staggered panels).
  - **Effect heterogeneity** across cohorts or time-since-treatment.
- Intuition: with homogeneous effects, TWFE implicitly “adjusts” the already-treated controls back to their counterfactual by subtracting the one constant τ. Under heterogeneity, no single adjustment works—so contaminated “controls” bleed into the estimate.

Staggered DID: the core idea

It is simple at bottom:

- You could imagine a DID at each relevant time point estimating the **cohort-time effect** $ATT(g, t)$: the ATT at time t for the cohort first treated at time g.
- If these are well formed (choosing appropriate controls each time) then it's just a question of how to average up the $ATT(g, t)$'s.
- Modern DID just points out that you need to construct these then *explicitly* weight them.

Simplest fallback: run a clean 2×2 DID for each adoption wave, look at them, average however you want.

Estimators for Staggered DID

The main approaches are:

Approach	**What it does**
Sun & Abraham (2021)	Event-study estimator using cohort-specific relative-time indicators; avoids contamination from already-treated units; valid under heterogeneous effects and staggered timing.
Callaway & Sant'Anna (2021)	Estimates group-time $ATT(g,t)$ using only not-yet-treated units as controls; flexible aggregation; avoids negative weighting.
Borusyak, Jaravel & Spiess (2021); Liu, Wang & Xu (2022)	Fits $Y(0)$ model with unit + time FEs on *untreated cells only*, impute $\hat{Y}_{it}(0)$ for treated cells, then aggregate effect estimates explicitly. Essentially g-computation with FE predictors. Implemented in `fect`.
de Chaisemartin & D'Haultfœuille (2020)	Reweights valid 2×2 DID comparisons with convex weights; interpretable under heterogeneity.

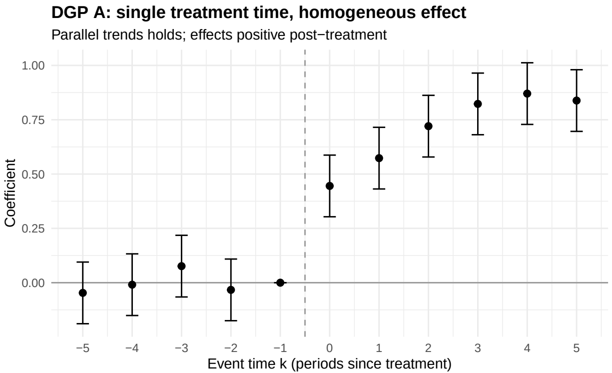
Event-study design: single treatment time

Setup. DGP A: half the units treated at one time, rest never-treated. *Parallel trends holds.* Post-treatment effect is positive, builds slowly.

Estimator. TWFE with leads and lags; $k = -1$ omitted as reference:

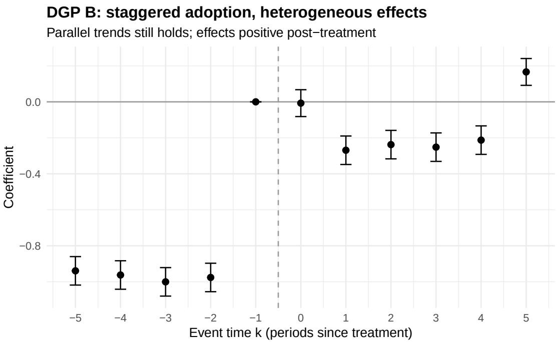
$$Y_{it} = \alpha_i + \gamma_t + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}\{t - E_i = k\} + \varepsilon_{it}$$

Use it to (a) visualize effect dynamics (β_k, $k \geq 0$); (b) check leads ($k < 0$) ≈ 0—a visual PT diagnostic.



Event-study design: staggered adoption

Setup. DGP B: three cohorts adopted at different times; effect heterogeneous across cohorts. *Parallel trends still holds.* Effects still positive. **Same TWFE event-study regression** as before.



- Leads appear strongly non-zero—a *false* PT alarm (PT actually holds).
- Lags flip sign, even though every true cohort-time effect is positive.
- Pathology is entirely from the estimator, not the DGP.
- Modern methods (SA, CS, BJS / `fect`) give event-study plots without it.

But remember two things:

Parallel trends is a strong assumption

- If time-varying influences might lead to treatment and changes in "growth", i.e. $Y_t(0) - Y_{t-1}(0)$, then parallel trends is violated.
- Modern DID methods don't touch this.
- Parallel trends (in the relevant period) is never testable. Don't be fooled by pre-trend tests.